

Marco Mo - Data Engineer

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SUMMARY

Data Engineer building Python and SQL pipelines for ingestion, transformation, validation, and API integration. Designs canonical data models, batch workflows, database services, data-quality checks, and Dockerized deployments.

EXPERIENCE

Junior AI Specialist

Jun 2026 - Sep 2026

ADC Nederland | Eindhoven, The Netherlands

- Developed four internal data and AI systems using Python, SQL, and APIs.
- Replaced 5-10 minutes of manual entry per order with reusable templates.
- Converted 7,000-8,000 orders to JSON in approximately 30 seconds.
- Preserved controlled review for 10-20% of orders missing production details.

Teaching Assistant

Sep 2021 - Jul 2025

Eindhoven University of Technology | Eindhoven, The Netherlands

- Supported students in Python, pandas, SQL, databases, and analytical debugging.
- Diagnosed data and code errors and guided reproducible solutions.

Software Technical Lead

Sep 2021 - Jul 2023

Student Team SimEnergy | Eindhoven University of Technology

- Integrated ESP32 ingestion with preprocessing, predictions, aggregation, and dashboard delivery.
- Coordinated data interfaces across software, hardware, and business requirements.
- Implemented hourly through yearly aggregation for appliance-level time-series data.

PROJECTS

Omnimap - Multi-Format Order Ingestion and Migration Platform

- Converted XML, JSON, CSV, and fixed-width files into validated REST API payloads.
- Built canonical schemas, versioned mappings, transformations, classifiers, and validation rules.
- Scanned 7,773 files, validated 6,044 conversions, and detected 1,104 duplicates.
- Added batch reporting, Pydantic models, Alembic migrations, tests, and Docker Compose.

MCP Server for the Wihabo Order Database

- Built FastMCP server exposing 67 typed tools over REST and MySQL.
- Added Pydantic schemas and aggregate SQL tools for structured access.
- Enforced tenant authentication, PII redaction, safe errors, and scoped caching.
- Supported JWT, remote transports, readiness checks, and concurrency controls.

quickSilver Graph Database - Engineering Data Systems

- Achieved fourth place based on combined computation-time and memory score.
- Implemented synopsis-based cardinality estimation and cost-based query planning.
- Optimized graph queries using CSR storage, caching, deduplication, and dynamic programming.

EDUCATION

MSc Data Science and Artificial Intelligence | Eindhoven University of Technology

2023 - 2025

- Thesis: Reward-Free Safe Reinforcement Learning Exploration Using Different Entropy Measures

BSc Data Science | Eindhoven University of Technology and Tilburg University

2020 - 2023

- Bachelor End Project: Health Platform Text Classification Using Active Learning

Honors Academy - AI Track | Eindhoven University of Technology

2021 - 2023

- Projects: time-series pipelines, ESP32 integration, aggregation, and visualization

SKILLS

Programming: Python, SQL, Bash, C++, Java

ETL & Processing: Apache Spark, ingestion, transformation, batch processing, API integration

Data Quality: canonical schemas, validation, deduplication, exception handling, lookup tables

Databases & Formats: PostgreSQL, MySQL, SQLite, ChromaDB, JSON, XML, CSV, Parquet

APIs, BI & DevOps: FastAPI, Pydantic, Alembic, Power BI, Docker Compose, CI/CD, pytest, Git

LANGUAGES

English (Fluent), Cantonese (Native), Mandarin (Fluent), Dutch (A2)